

Ben Bubnick

Master Career History

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Contents: Profile and skills (p. 1) | Professional experience (p. 1) | Teaching, research, and academic service (p. 7)

Additional work, service, leadership, and arts (p. 13) | Projects, publications, and speaking (p. 6) | Recognition, credentials, and education (p. 7)

Career Overview

Interdisciplinary data engineer, analyst, software developer, and educator with a foundation in physics and fine arts and more than a decade of experience building reliable data systems in healthcare, insurance, and color science. Leads ambiguous, high-stakes work from source discovery and requirements through modeling, automation, validation, migration, documentation, and production handoff, spanning cloud data platforms, more than 100 billion clinical records, multi-organization EHR integrations, revenue-cycle models, and applied machine learning. Known for turning complex technical and scientific problems into reproducible systems, clear explanations, and measurable delivery improvements.

Career experience also includes college physics and astronomy teaching, published astrophysics and printmaking research, independent product development and consulting, visual arts, community service, organizational leadership, and hands-on farm and operations work. Across these settings, combines analytical rigor, creative problem solving, practical craftsmanship, continuous learning, and a commitment to making knowledge, tools, and opportunities more accessible to others.

Skills and Domains

Programming Languages: *Current/Core:* SQL, Python, Ruby, Bash/Zsh; *Working Knowledge:* JavaScript, C#; *Historical/Academic:* VB.NET, C++, FORTRAN, Perl, IDL

Databases and Query Engines: *Current/Core:* Snowflake, PostgreSQL; *Working Knowledge:* Amazon Athena, Amazon Redshift, Azure Synapse Analytics, Microsoft SQL Server, MySQL; *Historical/Academic:* Oracle, Hive, Impala

Data Engineering and Modeling: *Current/Core:* dbt, SQLMesh, Snowpark, ETL/ELT development, data integration and transformation, dimensional modeling, data-quality testing, source-to-target mapping, reconciliation, data warehouses, data lakes; *Working Knowledge:* Spark, Kafka, Data Vault modeling; *Historical/Academic:* Hadoop/HDFS, Pig

Cloud Platforms: *Current/Core:* AWS (S3, Glue, Lambda, CloudWatch), Microsoft Azure (Data Factory, Functions, Blob Storage)

DevOps, Testing, and Automation: *Current/Core:* Git/GitHub, GitHub Actions, Docker, AWS CLI, CI/CD, automated testing, RSpec, RuboCop, Liquibase, Jinja, Make, CLI environment setup, Python virtual environments; *Working Knowledge:* Azure DevOps, Jenkins, Gradle, Postman, regular expressions, LINQ, RubyLLM, Tesseract OCR

Healthcare Data and Interoperability: *Current/Core:* clinical, payer, eligibility, revenue-cycle, and financial data; HL7 v2/v3, FHIR, EDI/837, EHR/EMR, HIPAA; *Working Knowledge:* HEDIS, HITRUST

Analytics, Machine Learning, and BI: *Current/Core:* data analysis, machine learning, predictive modeling, statistics, pandas, Jupyter, Power BI; *Working Knowledge:* Tableau, R; *Historical/Academic:* SAS/GRAPH

Scientific Computing: *Working Knowledge:* numerical analysis, mathematical modeling, simulations, linear algebra, error analysis; *Historical/Academic:* physics, astrophysics, stellar spectroscopy, photometry, optical measurement, radiative-transfer modeling, Fourier analysis, Monte Carlo methods, numerical interpolation, numerical integration and differentiation, IRAF, SpeXtool, Mathematica, MATLAB, LabVIEW, Maple, PDL, JMP

Teaching and Communication: *Current/Core:* technical writing and documentation, stakeholder communication, requirements facilitation, mentoring, public speaking, LaTeX; *Working Knowledge:* curriculum development, instructional design, research publication; *Historical/Academic:* physics and astronomy instruction, peer instruction, Just-in-Time Teaching, Interactive Lecture Demonstrations

Business and Creative Tools: *Current/Core:* Microsoft Office, bookkeeping, accounting, tax compliance, marketing, client acquisition, promotional-material development; *Working Knowledge:* Adobe Illustrator, Adobe Photoshop, GIMP; *Historical/Academic:* AutoCAD

Professional Experience

HHaExchange

Oct. 2025 - Jun. 2026

Healthcare Data Integration Consultant - Contract via Veersa Technologies

Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox

Led healthcare source analysis, data-quality investigation, and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, establishing reusable patterns for faster client onboarding.
- Implemented reusable SQL/Jinja transformation logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains.
- Created or materially redesigned 30 automated SQL/dbt data-quality tests covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.
- Translated ambiguous business requirements into documented mappings, acceptance criteria, onboarding standards, and 12 data-quality investigation runbooks.
- Conducted feasibility analysis of source-system structures and implementation constraints to shape integration scope, modeling strategy, and delivery approach.
- Structured target models using FHIR-like concepts where appropriate to improve consistency and extensibility for downstream integration and analytics.
- Defined surrogate-key, natural-key, and crosswalk strategies supporting stable identity resolution across related source systems.
- Resolved data-quality and mapping issues through targeted test-case analysis and review of failing source and target records.
- Collaborated with client and internal stakeholders to resolve source ambiguity, refine requirements, and translate business needs into production-ready data assets.
- Established reusable patterns for onboarding additional sources and extending integration workflows.

Adonis

Mar. 2025 - Jul. 2025

Healthcare Revenue Cycle Data Consultant - Contract via Health Data Movers*Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker, GitHub Actions*

Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding, analytics, and reporting.

- Built, documented, validated, and maintained 54 SQLMesh models that delivered standardized revenue-cycle datasets for a provider whose network served 4,598 families across 113 locations and processed 86,388 annual authorizations.
- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
- Diagnosed undocumented source-key versioning and implemented composite business grains with SHA-256 surrogate keys across 10 charge and payment models, eliminating duplicate-key failures.
- Authored implementation playbooks, technical handoff documentation, and a QA runbook for a repeatable healthcare-data validation workflow.
- Built a reusable SQLMesh/Snowflake ingestion and transformation framework using templated models, data contracts, CI checks, and promotion gates.
- Added schema-drift alerts, parity checks, reconciliation logs, deterministic rerun procedures, and incident-handling guidance for UNIX-like environments.
- Packaged a lightweight operations forecasting and alerting pattern as a tested Python and Docker service with GitHub Actions deployment automation.
- Standardized transformation logic across clients and translated business requirements into technical specifications to improve delivery predictability.

GrowData Analytics

Sep. 2024 - Oct. 2025

Independent Data Automation and Analytics Consultant*Technologies: Python, Ruby, RubyLLM, RSpec, RuboCop, Tesseract OCR, LLM APIs, Docker, AWS, GitHub Actions*

Built and operated an independent consultancy focused on converting semi-structured documents into validated, auditable data products.

- Built and validated a preproduction document-to-data MVP that converted semi-structured invoices into auditable transaction records, achieving 95% field-level accuracy across a stratified 500-invoice test suite.
- Instrumented the MVP with automated tests, audit logging, retry controls, and confidence routing, achieving 88% straight-through processing and under 1% system failure.
- Productized Tesseract OCR and rule-based post-processing as a small Python/Ruby service with containerized deployments and versioned configuration.
- Added schema checks, cross-field validation, reconciliation, idempotent reruns, fixtures, golden-file tests, linting, and release automation.
- Documented runbooks and failure modes covering timeouts, low-confidence OCR, and retries.
- Built a lightweight analytics layer tracking usage, error categories, latency, and throughput to guide feature tuning and cost control.
- Managed infrastructure as code for a minimal AWS footprint using S3, Lambda or scheduled jobs, and CloudWatch alerts through Bash/Zsh and Makefile workflows.
- Managed the business's bookkeeping, accounting, and tax obligations.
- Led marketing and client acquisition, including creating promotional materials for the business.

Arkos Health

Jun. 2023 - Jan. 2025

Senior Data Analyst

Technologies: AWS S3, Athena, Redshift, Glue, dbt, PySpark, Docker, Jupyter, Power BI, Tableau

Delivered healthcare analytics, recurring reporting, and data-quality investigations while improving the governed data foundation used by the BI team.

- Architected a greenfield AWS data lake integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets; automated CI/CD and targeted model refactoring accelerated deployments and improved query performance.
- Integrated Jupyter-based analytics and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring, executive, and BI reporting.
- Instrumented scheduled pipelines with monitoring, alerting, and more than 10 failure classifications; authored three AWS/dbt incident runbooks supporting rapid detection and recovery.
- Mentored engineers on code review, testing, and pipeline design and authored internal delivery standards.
- Implemented Jira and GitHub workflows to centralize task tracking and improve coordination across the analytics team.
- Expanded cross-departmental collaboration to bring operational context into analytics decisions.
- Mentored junior team members in code review, testing, pipeline design, and delivery practices.
- Led goal-setting sessions to establish shared objectives and a consistent tracking cadence.
- Initiated recurring stakeholder-feedback sessions to refine deliverables against business needs.

Merative

Jul. 2022 - Mar. 2023

Senior Data Scientist

Technologies: Microsoft Azure, Azure Data Factory, Blob Storage, Azure Functions, SQL Database, Synapse Analytics, Python, SQL, Azure DevOps

- Engineered a seven-layer Python and SQL migration-acceptance suite with zero unresolved discrepancies at sign-off across tested record counts, partitions, schemas, nulls, duplicates, field values, aggregates, and business rules.
- Led HIPAA-compliant deidentification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.
- Designed and implemented automated ETL testing and regression suites that detected schema drift and transformation errors early in the migration lifecycle.
- Validated end-to-end Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations during the transition from HDFS to Azure.
- Collaborated with multidisciplinary engineering teams to refine architectural designs and preserve workflow continuity and data fidelity.
- Documented test plans, platform-migration procedures, and onboarding guidance for future quality-assurance engineers.
- Enforced data-governance and security protocols supporting audit-risk reduction and HITRUST readiness.

IBM

Jun. 2015 - Jul. 2022

Advanced through five roles delivering healthcare analytics, data architecture, client integration, data-quality improvement, and large-scale migration validation.

Technologies: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, T-SQL, MySQL, PostgreSQL, PL/SQL, MQL, XPath, XQuery, Apache Pig, Jaql, JSONPath, Docker, Jenkins, Tableau, HL7

- Supported employee onboarding from 2016 to 2018, helping new colleagues learn team practices, systems, and technical workflows.
- Served on tiger teams supporting the HDFS-to-Azure data-platform upgrade and participated on the data-governance team.
- Participated in internal coding jams, hackathons, and cross-team delivery initiatives.
- Played a central role in the REACH cross-team training initiative, teaching SQL fundamentals to nontechnical colleagues and ETL, architecture, data-science, and predictive-modeling practices to technical colleagues.

Senior Data Scientist

Nov. 2021 - Jul. 2022

- Served as a lead member of Provider Analytics, integrating disparate and centralized sources into data marts and supporting data mining, statistical-model development, and advanced analytics.
- Worked with senior leaders and colleagues to identify opportunities, define objectives, and formulate analytics strategies that produced measurable impact.
- Served as technical lead across electronic health records, insurance claims, scheduling and financial data, social determinants of health, and other relevant sources.
- Developed migration-validation analysis spanning more than 150 billion deidentified health records, reconciling source and target schemas, partitions, values, aggregates, and business rules during the transition from legacy HDFS infrastructure to Azure.
- Provided production and configuration support for Phytel population-health applications, troubleshooting data and operational issues affecting provider configuration, care-gap identification, patient outreach, and quality-reporting workflows.
- Supported data and integration work for Pennsylvania's statewide P3N health-information exchange,

encompassing master patient identity, provider directories, clinical-document discovery, patient consent, and real-time ADT encounter notifications.

- Supported the controlled decommissioning of legacy Explorys capabilities by analyzing downstream dependencies and validating continuity across normalized clinical, claims, laboratory, billing, and operational datasets.
- Supported Cleveland Clinic Foundation's accelerated SuperMart migration from MySQL to Greenplum by assessing dashboard dependencies and defining the changes required to preserve reporting continuity.
- Created and delivered Phytel configuration training covering provider activation, system changes, support procedures, and the clinical-data workflows underlying proactive population-health management.
- Performed postmigration validation for an enterprise Jira transition, reconciling projects, issues, workflows, custom fields, permissions, histories, and relationships and documenting discrepancies for remediation before cutover.
- Developed and delivered peer-led Azure instruction through the REACH initiative, translating cloud architecture, storage, data integration, security, and analytics concepts into practical guidance for technical colleagues.
- Led delivery of a tailored analytics and reporting project using client administrative data.
- Engineered a Spark, Hive, and Impala analytics framework powering 50+ prior-authorization lag-time and approval-bottleneck reports with daily operational and periodic regulatory and financial snapshots.

Senior Data Architect

Dec. 2020 - Nov. 2021

- Served as a senior member of Data Onboarding & Governance, integrating healthcare data, enriching metadata, and configuring data models, analytics, reports, and visualizations.
- Served as a subject-matter expert to project data architects and delivery, analytics, and integration teams; directed configuration of the data-onboarding platform and supporting infrastructure.
- Led a cross-functional team as the clinical data analyst for the 2020/2021 measure-updates project.
- Created an exploratory machine-learning COVID-19 prototype to surface hospitalization-risk signals for retrospective analyst review.
- Developed a topic-modeling project using nearly six million semi-structured provider notes.
- Developed a surgical-scheduling optimizer and CNN/RNN case-card prediction proof of concept that was being evaluated as a potential new offering.
- Architected Spark and Hadoop pipelines integrating more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of measures and registries.
- Led a HIPAA patient-deidentification project drawing from more than 100 data sources for content analytics and measures.
- Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations, standardizing EDI batch schemas while enabling Python/Jupyter lineage tracing and end-to-end validation gates.
- Served in an advisory capacity for development of a new data platform.
- Closed project-documentation gaps by organizing material from more than 100 past projects.

Data Architect

Nov. 2017 - Dec. 2020

- Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing Jira-tracked validation rework by 25% and mapping rework by 45%.
- Led a multi-organization healthcare integration spanning 70+ EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules and coordinating delivery with engineering teams.
- Built the data-onboarding platform and associated infrastructure, remediated UAT data issues, and reported on data health during onboarding.
- Earned AI certification and peer-reviewed patient-matching algorithm tuning across the project portfolio.
- Curated 25 pages of AI-workflow documentation and mentored new hires in the AI-tuning process.
- Spearheaded a new-hire mentoring program that standardized technical ramp-up and prepared colleagues to contribute to billable client delivery.
- Received the Lab Services Award for coordinating internal project teams to deliver client needs on time.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.34%.

Software Developer, Data Platform

Sep. 2016 - Nov. 2017

- Engineered an HL7 v2/v3 parsing pipeline in Pig for approximately 30 internal users, replacing a Mirth-based workflow and eliminating \$30,000 in annual vendor costs.
- Rewrote and reorganized project documentation to streamline knowledge transfer.
- Redeveloped the employee education program to make new-hire training more efficient and repeatable.
- Received the Eminence and Excellence Award in 2017 for an analytics project.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.17%.

Data Scientist

Nov. 2015 - Sep. 2016

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Served as principal integration specialist on more than 60 projects for 20 clients across the broader 2015–2018 data-ingestion period.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to

standardize integration development and troubleshooting practices.

- Built a Python, Ruby, and SQL extraction tool that eliminated unnecessary story points and reduced average engineering hours per client integration.
- Eliminated an average of three weeks from employee onboarding.

Explorys, an IBM Company

Jun. 2015 - Nov. 2015

Data Science Consultant - Contract via Randstad USA

Technologies: Hadoop/HDFS, Ruby, Pig, React, Node.js, JavaScript, SQL, Oracle, Jenkins

Supported clinical-data integration and population analytics across more than 10 client data warehouses. IBM acquired Explorys in 2015; the overlap with the IBM tenure reflects the acquisition and transition rather than two unrelated concurrent roles.

- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.
- Identified and extracted data from heterogeneous client systems; led technical requirements gathering, discovery, documentation, and implementation; and coordinated acquisition, installation, and configuration of VPN hardware supporting client integrations.
- Developed a cron-based control for defining blackout periods around client data pulls.

AmTrust Financial Services

Mar. 2014 - Jun. 2015

Application Development Consultant - Contract via TEKsystems

Technologies: VB.NET, ASP.NET, C#, Web APIs, CSS, JavaScript, HTML5, SQL, PL/SQL, LINQ, R

Developed ISO-based web forms, multi-tier MVC components, Web APIs, and user interfaces for commercial insurance policy software.

- Designed, developed, tested, and maintained full-stack commercial-insurance applications spanning VB.NET and ASP.NET interfaces, multi-tier components, Web APIs, and SQL data services.
- Implemented a new line of business in VB.NET and Microsoft SQL Server, served as principal developer for a new product offering, and contributed to two major software-architecture initiatives in 2015.
- Evaluated the advantages, risks, and implementation costs of requested features and recommended maintainable solutions grounded in object-oriented design principles.
- Served as the working liaison between business analysts and developers, translating requirements, reporting progress, and helping prioritize delivery needs.
- Coordinated developer schedules and troubleshoot production problems across multiple application environments.
- Created a unit- and integration-test suite for multi-tier MVC components, improving regression coverage and establishing stronger team testing practices.
- Developed more than 65 ISO rating-based web forms and analyzed page timings in R to identify optimization targets and improve application performance.
- Refactored and maintained the existing VB.NET insurance-rating codebase to improve readability, performance, and long-term maintainability.
- Supported the technical integration of acquired businesses into the commercial policy platform.

PPG Industries

Apr. 2012 - Aug. 2013

Color Scientist

Technologies: VB.NET, C#, SQL, C++, FORTRAN, JMP, spectrophotometry, numerical methods

Developed software and radiative-transfer models for automotive paint formulation, spectroscopic profile analysis, and technology-risk assessment.

- Improved color-matching classification accuracy using numerical integration methods.
- Built cross-instrument correlation software in VB.NET and JMP to improve measurement consistency and support statistical risk analysis.
- Debugged a VB.NET and FORTRAN optimization routine for a numerical radiative-transfer solution.
- Wrote a Visual C++ and VB.NET DLL wrapper for a loaned spectroscopic instrument and developed formulation methods for an automated color-matching facility.
- Led cross-functional color-science and marketing initiatives from concept through on-time, in-full delivery for the Automotive Refinish business.
- Served as business owner for color data-management software, directing its global development, launch, training, adoption, and application.
- Managed relationships with third-party equipment and software suppliers, acting as the primary contact for technical coordination and issue resolution.
- Managed a portfolio of color-related intellectual property and supported the translation of research into operational tools and practices.
- Served as the global liaison among Color, Technical, Operations, and IT teams, translating color-science requirements into consistent systems, workflows, and communications.
- Designed and standardized color workflows and best practices for internal operations and external customers, leading the change-management efforts required for adoption.
- Evaluated color processes, instrumentation, and data flows to identify technical and operational risks and develop practical mitigation strategies.

- Developed statistical models from established theory, experimental data, and novel approaches to improve color measurement, matching, and process decisions.
- Championed a more science-centered organizational approach to color by introducing analytical methods, shared standards, and evidence-based decision-making.
- Informed marketing strategy and communications by connecting customer needs with technical capabilities and broad operational insight.

Selected Independent Projects

iphone-cam-lite	2025 - 2026
Native Swift/SwiftUI macOS menu bar application that selects an iPhone Continuity Camera and microphone in Zoom or Microsoft Teams, then reads back the application state to verify that the requested devices were actually selected. Features modular application adapters, permission-aware onboarding, system diagnostics, configurable device and accessibility labels, persistent settings, actionable error reporting, hermetic Swift tests, continuous integration, and scripted application packaging.	
farm-invoice-parser	2025
Ruby command line prototype that converts handwritten or printed farm invoices into structured transaction records. The pipeline extracts text from invoice images with Tesseract OCR, sends the result to configurable language-model providers through RubyLLM, parses the response as JSON, and validates it against a versioned invoice schema. It also provides an OCR-only diagnostic mode and includes RSpec, RuboCop, and GitHub Actions scaffolding.	
cfid_snow_fence	2025
PPython applied modeling experiment for evaluating potential snow fence locations using measured wind data. The project maps wind vectors onto a two-dimensional spatial grid, estimates pressure fields through a finite difference Navier–Stokes/Poisson calculation, and aggregates the resulting simulations. It then trains an XGBoost classifier using position, wind speed, and pressure to identify low wind regions as a proxy for potential snow drift accumulation.	
claims_case_study	2023
End to end healthcare workforce planning case study analyzing Cleveland area clinician, facility, and claims data for 2021–2022. The project evaluates source data quality, converts the original workbook into schema preserving Parquet datasets, standardizes the analytical data, and performs exploratory and time series analysis. ARIMA forecasts with error diagnostics project 2023 demand for Internal Medicine, Urology, and Anesthesiology, culminating in a written technical report and stakeholder oriented presentation.	
HL7-ingestion	2023
Java 21 Spring Boot/ Jersey service that accepts HL7 v2 messages as multipart file uploads or plain text and converts them into JSON encoded FHIR R4 collection Bundles using the LinuxForHealth converter. The project packages the service with Docker and a Helm chart for Kubernetes deployment, includes GitHub Actions automation, and exercises transformation and error handling through tests that validate generated Patient, Encounter, Observation, Practitioner, and AllergyIntolerance resources.	
quality_measures_engine	2023
Python/PyMongo prototype for defining and evaluating population health quality measures from normalized patient records. It represents measure metadata, parameters, calculated dates, initial populations, denominators, numerators, exceptions, and exclusions as reusable JSON definitions, then translates those definitions into MongoDB queries or map reduce operations. Fixtures and automated tests demonstrate the design using sample NQF cervical cancer screening and pneumonia vaccination measures.	
DETL-design-challenge	2023
Architectural design for a dynamic ETL system that transforms heterogeneous clinical research data into the OMOP Common Data Model. The design combines detailed source to target specifications, human readable transformation rules, generated SQL, and explicit review, testing, and debugging stages so that common mappings can be automated without removing expert oversight from complex semantic decisions. It also documents data quality controls, terminology mapping, de-identification considerations, and implementation options involving Apache NiFi, Azure Data Factory, and Python.	
zams_cluster_candidacy	2019 - 2020
Python scientific computing pipeline for evaluating stellar cluster candidacy from Washington system photometry. The analysis filters observations by photometric uncertainty and cluster radius, incorporates decontamination membership probabilities, and corrects color–magnitude diagrams for reddening and distance. It then traces candidate cluster sequences with kernel density contours and compares them with zero age main sequence and isochrone models across age and metallicity ranges, producing per cluster and aggregate diagnostic plots.	
gamma_rays	2019
Python and Jupyter machine learning research project for reconstructing events from high energy gamma ray telescope data. It reads simulated FITS event tables, balances gamma ray and hadronic background samples, and compares random forest, multilayer perceptron, and exploratory neural network classification approaches using confusion matrices, ROC/AUC, Q-factor, and calibration metrics. The project also trains regression models for event energy and direction reconstruction and includes reusable prediction code, persisted models, technical reports, and disseminated notebook analyses.	

Star-Chart

2015

Browser based HTML, CSS, and JavaScript astronomy application that renders the visible night sky for a selected location and date. It accepts browser geolocation or manually entered coordinates, calculates Julian date and local sidereal time, projects catalog stars according to right ascension, declination, and magnitude, and computes positions for the Sun, Moon, and visible planets. Stars, planets, and optional constellation lines are drawn on an interactive, zoomable canvas.

Professional Recognition**Special Recognition, Health Data Movers - Client Delivery**

2025

Recognized for diagnosing undocumented source-key versioning in a revenue-cycle integration and implementing composite business grains with deterministic SHA-256 surrogate keys across 10 charge and payment models, eliminating duplicate-key failures and stabilizing client delivery.

Arkos Health BI Star

2023

Internal Business Intelligence recognition received during a period marked by establishing the team's GitHub workflow, delivering 27 BI reports, creating the Arkos Power BI theme and visual standards, and strengthening PHI deidentification and analytics documentation supporting value-based-care operations.

Manager Recognition for Transition Leadership, Merative - Systems Migration

2022

Recognized by management for transition leadership while assuring the HDFS-to-Azure platform migration: built a seven-layer Python and SQL acceptance suite, validated Azure Data Factory, Blob Storage, and Synapse workflows, and led HIPAA-compliant deidentification testing across data representing approximately 300 million patient lives, reaching sign-off with zero unresolved discrepancies.

IBM Champion Learner

2016 - 2022

Progressed from Bronze in 2016 to Silver in 2017, then earned Gold annually from 2018 through 2022 through IBM Your Learning continuing education, completing 1,477 learning hours and earning 43 credentials. Gold recognition required at least 200 annual learning hours and three new badges or certifications.

Lab Services Award, IBM - Cross-Team Delivery Coordination

2018

Recognized for leading a multi-organization healthcare integration spanning more than 70 EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules, and coordinating engineering and delivery teams to meet client commitments.

Eminence and Excellence Award, IBM - Data-Curation Tool Development

2017

Recognized for engineering an Apache Pig pipeline that parsed HL7 v2 and v3 data for approximately 30 internal users, replacing a Mirth-based workflow and eliminating \$30,000 in annual vendor costs.

Color Lab Automation Award, PPG Industries - Expanded Reporting

2012

Recognized as part of the Color R&D team supporting PPG's new computer-controlled automotive-refinish color-matching laboratory in Cleveland; expanded automated data collection and analysis reporting used to evaluate pigments and increase formulation throughput, precision, and operational efficiency.

Mary J. Hanna Research Fellowship, University of Cincinnati

2010

Department of Physics fellowship providing financial support for completion of M.S. research using near-infrared photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk star clusters.

Academic Scholarships and Honors

2000 - 2010

Received University Graduate Scholarships supporting physics study from 2008 to 2010; earned University of Cincinnati Dean's List recognition from 2005 to 2007; received Ohio University's faculty-endorsed Provost's Undergraduate Research Fund grant for the 2003 low-toxic photo-etch project and earned Dean's List recognition from 2002 to 2004; received the Ohio Proficiency Scholarship in 2000 after passing all five sections of the state's twelfth-grade proficiency examination.

Teaching and Academic Research

Across these institutions, used interactive, peer-led, and just-in-time teaching methods.

Lorain County Community College

Aug. 2016 - May 2019

Adjunct Professor - Taught the laboratory component of algebra-based General Physics I (then PHYC 151) and the three-credit ASTY 152 Solar System Astronomy lecture course.

- Guided students through laboratory work in mechanics, energy, momentum, rotation, and thermal physics, emphasizing measurement uncertainty, quantitative analysis, and scientific reporting.
- Developed and taught online astronomy instruction covering astronomical observation, planetary-system formation, terrestrial and Jovian planets, small bodies, and space exploration.
- Exceeded institutional student-evaluation benchmarks by an average of 16% in 2017 and 17% in 2018.
- Reduced reported student confusion by 28% by reorganizing online instructional materials and reducing the need for clarification during office hours.
- Developed and published more than 70 hours of publicly available recorded lectures and laboratory content to support flexible, on-demand learning.
- Used Just-in-Time Teaching strategies to adapt instruction for underperforming students based on pre-class responses.

Cincinnati State Technical and Community College

Feb. 2011 - Jun. 2011

Adjunct Professor - Developed and delivered the integrated Technical Physics I–III lecture and laboratory sequence for technical-program students with varied mathematics preparation.

- Delivered two weekly lecture hours and three laboratory hours per course across applied mechanics, thermodynamics, electricity and circuits, waves, and optics.
- Wrote laboratory instructions and coordinated team-based measurement, data-analysis, and technical-report work.

Miami University

Aug. 2010 - Jun. 2011

Adjunct Professor - Developed and taught three-credit general-education courses PHY 101 Physics and Society and PHY 111 Astronomy and Space Physics.

- Designed a concept-first applied-physics curriculum for non-science majors, translating advanced technical subjects into accessible, decision-ready frameworks without relying on calculus.
- Taught students to use order-of-magnitude estimates, energy density, power, conservation laws, and comparative risk to evaluate claims involving transportation fuels, electricity generation, climate, and emerging technologies.
- Connected nuclear physics and radioactivity with power generation, medical imaging, radiation risk, and nuclear security; linked gravity, waves, and the electromagnetic spectrum with satellites, GPS, radar, communications, and remote sensing.
- Structured lessons around consequential real-world questions, teaching students to distinguish scientific constraints from policy and value judgments and to communicate evidence-based conclusions to nontechnical audiences.
- Taught astronomy topics spanning planetary systems, stellar evolution, galaxies, cosmology, and the scientific search for extraterrestrial life.

University of Cincinnati

Jan. 2006 - Sep. 2010

Held concurrent research and teaching appointments in physics, astronomy, and engineering computation.

Research Assistant, Department of Physics

Sep. 2006 - Sep. 2010

Conducted near-infrared astronomical research using observations from the 0.8–5.4 micrometer SpeX spectrograph at the NASA Infrared Telescope Facility.

- Reduced and analyzed 0.8–2.4 micrometer, $R \sim 2000$ SpeX spectra and measured diagnostic H I, He I, He II, C IV, N III, and CO spectral features across nine candidate cluster members.
- Co-developed an astrometrically calibrated catalog of 130 sources by registering SpeX acquisition images to 2MASS; compiled available *JHK* photometry and used DAOPHOT to separate blended sources in the crowded field.
- Helped classify the cluster's massive stellar population, identifying an O5 V dwarf, several early-B stars, and a massive young stellar object exhibiting CO band-head emission from dense circumstellar material.
- Combined spectral classifications with 2MASS photometry to estimate cluster extinction ($A_V \sim 21\text{--}23$), distance (1.0–1.75 kpc), age of no more than a few million years, and mass of approximately 10^3 solar masses.
- Cross-matched infrared and radio-source positions and identified candidate near-infrared counterparts for three deeply embedded radio sources associated with early star formation.
- Published the resulting 130-source astrometric and photometric catalog through the international VizieR astronomical data service.
- Acquired follow-up L-band spectra of Galactic OB stars with SpeX during three consecutive nights at NASA's IRTF.
- Estimated stellar radial velocities by measuring pixel-space shifts of CO absorption features against normalized standard-star spectra.
- Performed astronomical spectroscopy, photometry, classification, calibration, signal analysis, and noise reduction using IRAF, IDL, SpeXtool, Mathematica, and Perl.

Teaching Assistant

Jan. 2006 - Jun. 2010

Led approximately 60 laboratory meetings for more than 100 engineering, science, and preprofessional students in the algebra/trigonometry-based College Physics sequence (PHYS 111–113) and calculus-based General Physics sequence (PHYS 211–213).

- Taught experimental design, measurement uncertainty, data acquisition, graphical analysis, and technical reporting using PASCO DataStudio and Microsoft Excel.
- Served as teaching assistant and grader for the corresponding College Physics lecture sequence (PHYS 101–103) and General Physics lecture sequence (PHYS 201–203).
- Supported Astronomy and the Natural Universe lectures (PHYS 121–122) and astronomy laboratories (PHYS 125–127) through grading and instructional assistance.
- Designed interactive laboratory instruction using Peer Instruction and Interactive Lecture Demonstrations to identify and address student misconceptions in real time.
- Supported group recitations in freshman engineering physics through demonstrations, short lectures, and individual assistance.

Engineering Teaching Assistant

Oct. 2009 - Dec. 2009

Prepared and delivered Mathematica instruction for Numerical Methods for Engineering Design, an upper-division aerospace-engineering course.

- Developed step-by-step demonstrations applying symbolic and numerical computing to interpolation and approximation, numerical integration, nonlinear equation solving, systems of equations, and ordinary differential equations.

- Connected computational methods with aerospace analysis and design problems, showing students how to translate mathematical formulations into executable workflows and interpret the resulting output.
- Reinforced numerical accuracy, error, stability, and method selection so students could evaluate the credibility of computed engineering results.

Ohio University

Oct. 2003 - May 2004

Student Researcher, Fine Arts - Earned a competitive Provost's Undergraduate Research Fund award to adapt a wintgreen-oil toner-transfer process from lithography to intaglio photo-etch printing.

- Developed and tested a complete copper- and zinc-plate workflow spanning surface preparation, asphaltum grounding, high-contrast toner transfer, burnishing, clearing, aquatint, and staged etching.
- Replaced xylene- and acetone-based photo-transfer steps with methyl salicylate, eliminated the need for specialized exposure equipment, and demonstrated a process using less than \$20 in materials.
- Characterized critical process variables and failure modes, including ground thickness and cure time, solvent saturation, image resolution, transfer pressure, evaporation time, and temperature control during aquatint fusion.

Publications and Research Outputs**Applied Research and Technical Writing**

Bubnick, Ben. (2025). "Reproducible Postgres load of Tuva seed datasets." Internal applied research series, GrowData Analytics/HSEA. Document tracking no. 20250401-A. Internally published by HSEA, Aug. 2025.

Author's version, Apr. 1, 2025.

Bubnick, Ben. (2020). "Surgical Scheduling Optimization and Procedure Duration Prediction." Internal project report, IBM. Document tracking no. RC25712.

Bubnick, Ben. (2017). "Process and Requirements for CCD Data Integration." Internal white paper, IBM. Document tracking no. RC25043.

Bubnick, Ben. (2013). "Characterization of Surface Coatings Using RT Models for Color Matching." Unfiled patent application authored for PPG Industries and retained by the company as a trade secret. Ref. 1284 - February 2013.

Bubnick, Ben. (2010). "The B Stars of Westerlund 1." Unpublished internal stellar-spectroscopy analysis, University of Cincinnati.

Journal Articles and Data Catalogs

Bubnick, Ben. (2015). *AI paint formula prediction from spectroscopic measurements* (Version 1.2) [Interactive web application]. shinyapps.io.

Bubnick, Ben. (2010). *Massive stellar clusters in the disk of the Milky Way Galaxy* [Master's thesis, University of Cincinnati]. OhioLINK Electronic Theses and Dissertations Center. OhioLINK record

Hanson, M. M., & Bubnick, Ben. (2008). *JHK photometry in cluster [BDS2003] 107* [Data set]. VizieR Online Data Catalog (Catalog J/PASP/120/150). <https://doi.org/10.26093/cds/vizie.61200150>

Hanson, M. M., & Bubnick, Ben. (2008). Near-infrared spectroscopy of candidate members of the Galactic cluster [BDS2003] 107. *Publications of the Astronomical Society of the Pacific*, 120(864), 150–159.

<https://doi.org/10.1086/527403>

Bubnick, Ben. "A Novel Method for Low-Toxic Photo-Etch Printing." *MAPC Journal*, vol. 11, no. 2, Fall/Winter 2003, pp. 12-13.

Speaking Engagements**Surgical Scheduling Optimization and Procedure Duration Prediction - IBM AI Hackathon - Dec. 2020**

Presented a prototype that combined historical operating-room utilization and the current surgical wait list to optimize scheduling and improve total procedure-time estimates beyond traditional fixed-ratio methods; outlined CNN/RNN case-card prediction as a path toward a potential new offering.

IBM Watson Care Insights - IBM REACH Initiative - 2019

Delivered conceptual and hands-on training with synthetic patient data, explaining how longitudinal records, real-time HL7 feeds, and FHIR configuration support clinical-data integration and the delivery of insights within physicians' EHR workflows.

Machine Learning with Real-World Health Data - Content and Data Analytics Learning Exchange - 2018

Introduced an implementation-oriented approach to applying machine-learning algorithms to real-world health data, connecting clinical-data preparation, model development, evaluation, and interpretation.

Natural-Language Processing for Unstructured Data Lakes - Innovation Research Interchange Fall Networks Conference - Sep. 2018

Presented a healthcare case study using natural-language processing and topic modeling to turn nearly six million semi-structured provider notes in a data lake into analyzable themes and features.

Hadoop Ingestion and Data-Lake Concepts - Cross-Team Training Series - 2016

Designed and led a seven-part series connecting hands-on Hadoop/HDFS ingestion workflows with distributed storage, transformation, validation, troubleshooting, and the use of data lakes for heterogeneous clinical data.

Massive Stellar Clusters in the Disk of the Milky Way Galaxy - M.S. Physics Thesis Defense, University of Cincinnati - Jun. 2010

Defended research using near-infrared broad-band photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk clusters and estimate their distance, age, and approximate mass.

Astronomical Research Using 2MASS Data - 2MASS Research Poster Session - 2008

Presented results from combining 2MASS *JHK* photometry with IRTF/SpEx observations to characterize candidate members of the young Galactic cluster [BDS2003] 107, including its stellar population, extinction, distance, age, and approximate mass.

A Novel Method for Low-Toxic Photo-Etch Printing - Ohio University Research Poster Session - 2004

Presented Provost's Undergraduate Research Fund-supported work adapting a wintergreen-oil toner-transfer process from lithography to intaglio photo-etch printing, including a low-cost copper- and zinc-plate workflow that replaced xylene- and acetone-based transfer steps and eliminated specialized exposure equipment.

Creative and Visual Arts

Media: Relief printmaking (woodcut and linocut), intaglio (etching, photo-print, and aquatint), screenprint, and papermaking; oil and acrylic painting; graphite, ink, and mixed-media drawing.

Selected Exhibitions

Drawn and Quartered VII, HEDGE Gallery, 78th Street Studios, Cleveland, Ohio May 15–Jun. 5, 2015

Exhibiting Artist - Participated in a group exhibition of drawings. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

13th Annual Holiday Store, Heights Arts, Cleveland Heights, Ohio Nov. 6–Dec. 31, 2014

Exhibiting Artist - Presented relief prints and drawings. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Holiday Off The Wall Members Show and Sale, Zygot Press, Cleveland, Ohio Dec. 6–7, 2014

Exhibiting Artist - Presented printmaking work in the annual members' exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Waterloo Arts Fest, Collinwood, Cleveland, Ohio Jun. 29, 2013

Exhibiting Artist and Printmaker - Presented prints and participated in live printing with Zygot Press. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Tremont ArtWalk, Tremont, Cleveland, Ohio 2013

Exhibiting Artist - Participated in a group exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Holiday Off-the-Wall Exhibition, Zygot Press, Cleveland, Ohio Dec. 1–2, 2012

Exhibiting Artist - Presented printmaking work in the annual members' exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Juried Exhibition, Zygot Press, Cleveland, Ohio Jun. 1–Jul. 7, 2012

Exhibiting Artist - Presented printmaking work in a juried exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

11th Annual Holiday Store, Heights Arts, Cleveland Heights, Ohio Nov. 2–Dec. 30, 2012

Exhibiting Artist - Presented drawings. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Neighborhood Artists, Kennedy Heights Arts Center, Cincinnati, Ohio 2010

Exhibiting Artist - Participated in a group exhibition of paintings. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Final Friday Open Studios, Pendleton Art Center, Cincinnati, Ohio 2007 - 2008

Studio Artist - Presented paintings and drawings from a shared studio during recurring open-studio events. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

B.F.A. Thesis Exhibition, Jennings House Gallery, Ohio University May 2004

Exhibiting Artist - Presented senior printmaking work in Athens, Ohio. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Art Vs. Sound II, Lacuna Gallery, Athens, Ohio May 2004

Exhibiting Artist - Displayed original artwork in a local gallery exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Disaster, Lacuna Gallery, Athens, Ohio May 2004

Exhibiting Artist - Displayed original artwork in a local gallery exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Grand Opening, Lacuna Gallery, Athens, Ohio Apr. 2004

Exhibiting Artist - Displayed original artwork in the gallery's opening exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

No Joke, Lacuna Gallery, Athens, Ohio Apr. 2004

Exhibiting Artist - Displayed original artwork in a local gallery exhibition. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Art v. Sound, Lacuna Gallery, Athens, Ohio Apr. 1, 2004

Exhibiting Artist - Participated in a multimedia event featuring visual art and live DJ performances. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Undergraduate Juried Exhibition, Ohio University Art Galleries 2004

Exhibiting Artist - Presented work in the university's juried exhibition in Athens, Ohio. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Student Print Exchange, Ohio University Art Galleries Feb. 2004
Organizer and Exhibiting Printmaker - Organized an exhibition and print exchange for student printmakers in Athens, Ohio. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted/exchanged]; [edition and artist's-proof details, if applicable]).

Undergraduate Juried Exhibition, Ohio University Art Galleries 2003
Exhibiting Artist - Presented work in the university's juried exhibition in Athens, Ohio. **Works:** *[Artwork title(s)]* ([medium/media], [creation year(s)]; [sold/retained/gifted]; [edition and artist's-proof details, if applicable]).

Print Exchanges and Conferences

Mid America Print Council Conference Print City: Detroit 2014, Wayne State University, Detroit, Michigan Sep. 24–27, 2014
 Contributed work to an exchange portfolio. **Work:** *[Artwork title]* ([medium], [creation year]; edition [size]; [disposition of other impressions]; artist's proof [retained/not retained]).

Southern Graphics Council Conference Making Histories: Revolution and Representation, Boston, Massachusetts Apr. 2–5, 2003
 Contributed work to a student exchange portfolio. **Work:** *[Artwork title]* ([medium], [creation year]; edition [size]; [disposition of other impressions]; artist's proof [retained/not retained]).

Ohio University School of Art Student Print Exchange 2003
 Participated in a student print exchange organized by the Printmaking area. **Work:** *[Artwork title]* ([medium], [creation year]; edition [size]; [disposition of other impressions]; artist's proof [retained/not retained]).

Studio Access, Workshops, and Visiting-Artist Projects

Zygote Press, Cleveland, Ohio 2011 - 2014
 Completed independent studio work in etching, relief, and screenprint.

Morgan Art of Papermaking Conservatory, Cleveland, Ohio 2012
 Completed a papermaking workshop.

Manifest Drawing Center, Cincinnati, Ohio 2009 - 2010
 Participated in open figure-drawing sessions.

Art Academy of Cincinnati Community Education 2005
 Completed oil-painting instruction in Cincinnati, Ohio.

Debra Fisher Visiting-Artist Print Project, Ohio University 2003
Studio Assistant - Supported plate preparation, proofing, and editioning.

Dan Murphy Visiting-Artist Print Project, Ohio University 2003
Studio Assistant - Supported plate preparation, proofing, and editioning.

Ohio University School of Art Commission 2003
Commissioned Artist - Created *Tree on Paper*, an original print commissioned by Raymond Tymas-Jones, Dean of the College of Fine Arts.

Volunteer and Community Experience

HSEA Feb. 2025 - May 2025
ETL Brigade Volunteer - Built a reusable PostgreSQL data model and reproducible loading process for Tuva seed datasets, enabling community-health organizations to prototype analytics using claims, visits, and eligibility data. Added schema setup, normalization, and validation components to make the data consistently loadable and testable.

CSforAll Oct. 2018 - May 2019
Instructor - Supported computer-science instruction for Cleveland Metropolitan School District high-school students as part of a districtwide effort to make rigorous computer-science coursework available in every high school. Helped broaden access for students from underrepresented and low-income populations, with particular attention to learners without reliable computer or Internet access at home.

HIT in the CLE Mar. 2017 - Mar. 2019
Coach - Coached five-student high-school teams through a data-science competition, helping participants analyze a large dataset, develop evidence-based conclusions, and communicate their findings. Introduced students to health-information-technology careers while building data-analysis and presentation skills and supporting a more diverse Northeast Ohio health-IT talent pipeline.

Code for America Jan. 2015 - Jun. 2015
ETL Guild Volunteer - Applied ETL and data-science methods to initiatives that made difficult-to-use public records more accessible and machine-readable for research, planning, and civic use. Supported broader efforts to improve government transparency and help residents engage with public information.

St. Herman House - FOCUS Cleveland, Cleveland, Ohio 2013 - 2014
Daily Meal Program Volunteer - Supported a year-round meal program providing three meals a day to neighbors in need, helping deliver welcoming, dignity-centered service.
 • Distributed meal tickets, plated food, handed meals to guests, and served water and desserts in the dining room.

Heights Observer, Cleveland Heights, Ohio Feb. 2012 - May 2014
Volunteer Editor - Edited resident-submitted reporting for a nonprofit, citizen-produced newspaper serving Cleveland Heights and University Heights, improving accuracy and clarity while preserving each contributor's voice. Also contributed local arts and culture coverage, supporting civic participation, public discourse, and independent community journalism.

The News Record, University of Cincinnati

Feb. 2008 - Sep. 2010

Editorial Contributor - Wrote and edited campus content for the university's independent, student-run newspaper. Helped inform and educate the university community, amplify student perspectives, and maintain accurate, timely coverage of campus life.

TCP World Academy

Sep. 2008 - May 2009

Tutor - Tutored students in support of TCP World Academy's mission to develop academically engaged, independent learners. Reinforced classroom learning through technology and participatory micro-society activities that connected students' education with meaningful participation in a global society.

Project Astro at Ethel M Taylor Academy

2008

Volunteer Astronomer - Worked with an educator at Ethel M Taylor Academy to provide hands-on, inquiry-based astronomy and physical-science activities. Helped students investigate scientific questions and engage with astronomy as practicing scientists through the national Project ASTRO educator-astronomer partnership model.

Sigma Theta Epsilon, Beta Gamma Chapter, Cincinnati, Ohio

2006 - 2007

Chaplain - Provided spiritual leadership for the University of Cincinnati chapter of a national interdenominational Christian service fraternity. Supported fellowship, reflection, accountability, and service-oriented activities consistent with the fraternity's emphasis on Christian faith and service.

- Served as a chapter officer supporting the chapter's spiritual life and service mission.

Reach Out on Campus, Ohio University, Athens, Ohio

Sep. 2000 - Jun. 2004

Student Leader and Volunteer - Volunteered throughout undergraduate study and served as a Student Leader during the 2003–2004 academic year. Supported a Christian campus ministry focused on belonging, spiritual growth, leadership development, campus outreach, and service to others.

- Helped sustain student fellowship, Bible studies, outreach, and service activities across the Ohio University community and southeastern Ohio.

Leadership, Memberships, and Activities**Detroit Color Council**

2012 - 2013

Member of a nonprofit industry forum promoting communication and education among engineers, designers, scientists, and other professionals involved in automotive color and appearance. The council advances shared technical knowledge and practices intended to improve the quality of colored and decorative materials.

Zygote Press, Cleveland, Ohio

2011 - 2014

Member and independent studio artist working in etching, relief, and screenprint. Zygote is a nonprofit community printmaking studio, gallery, archive, and education center that expands artists' access to affordable professional workspace, instruction, exhibitions, residencies, and collaborative development.

Mid America Print Council

2003 - 2004

Student member of an educational and community-based organization that supports traditional and contemporary printmaking through conferences, exhibitions, publications, professional exchange, and research. Its stated values emphasize inclusion, diversity, integrity, and creative expression.

Southern Graphics Council (now SGC International)

2002 - 2004

Student member of an educational nonprofit that advances printmaking and graphic arts through professional exchange, exhibitions, publications, conferences, and recognition of artists. The organization promotes research, public appreciation, creative dialogue, and participation across diverse communities.

American Institute of Aeronautics and Astronautics Cincinnati Student Chapter

2009 - 2011

Member of a student chapter connecting participants with the aerospace profession and AIAA's emphasis on innovation, technical excellence, and professional collaboration. Participated in Students for the Exploration and Development of Space in 2009, supporting its mission to advance student involvement in space exploration through educational and engineering projects.

- Vice President, 2010.

University of Cincinnati Astronomy Club

2008 - 2010

Founding member in 2008 who helped establish a student community intended to cultivate interest in astronomy through observation, education, and shared exploration of the night sky.

- Secretary, 2009–2010.
- Treasurer, 2008–2009.

Additional Academic Activities

2006 - 2010

Participated in the Large Synoptic Survey Telescope seminar group, UC SPARKS, and University of Cincinnati physics departmental seminars, which provided forums for research discussion and scientific exchange. Also participated in the University of Cincinnati Mountaineering Club, which promotes safe wilderness skills, outdoor appreciation, stewardship, and community through instruction and group outings.

Society of Physics Students Cincinnati Chapter

2005 - 2008

Helped lead a chapter of the national professional association for physics students, which builds supportive scientific communities and promotes education, research, professional development, outreach, and public understanding of physics.

- Treasurer, 2006–2007.
- Secretary, 2005–2006.

Sigma Theta Epsilon Beta Gamma Chapter

2006 - 2007

Member of an interdenominational Christian service fraternity organized around faith, fellowship, service, and meeting unmet human needs.

- Chaplain, 2006–2007 academic year.

Athens Wargamers

2001 - 2004

Member of an Ohio University student community centered on tabletop wargaming, strategic problem solving, and social play.

- Treasurer, 2002–2003 academic year.

Academic Services**University of Cincinnati Geology Math Physics Library**

Mar. 2006 - Sep. 2008

Student Librarian - Supported research and collection operations for a specialized university library.

- Assisted students, faculty, and other patrons with research and reference needs and helped locate relevant materials.
- Maintained Library of Congress-classified collections, conducted inventory, and investigated missing items.

University of Cincinnati Learning Communities Program

Jun. 2006 - Aug. 2008

Peer Leader - Facilitated a learning community for approximately 25 first-year students co-enrolled in multiple courses, supporting their academic, social, and professional transition to university life.

- Prepared and facilitated twice-weekly meetings on course assignments, study strategies, time management, academic expectations, campus resources, and first-year transition needs.
- Provided individual and small-group mentoring, helping students form study networks, navigate university systems, and connect with faculty, advisers, and academic-support services.
- Planned community-building activities and coordinated with faculty, advisers, program staff, and campus organizations to strengthen belonging, peer accountability, and academic engagement.

Private Tutoring

Jun. 2006 - Aug. 2007

Private Tutor - Provided sustained one-on-one instruction to a student completing the calculus-based introductory physics sequence, PHYS 201–203.

- Adapted short lectures, demonstrations, and representative worked problems to the student's individual learning needs and progress across the three-course sequence.
- Taught a repeatable problem-solving framework built around diagrams, governing physical principles, dimensional analysis, estimation, and interpretation of calculated results.
- Diagnosed misconceptions through guided questioning and targeted practice, reinforcing difficult concepts with step-by-step explanations and alternative solution strategies.

University of Cincinnati

Summer 2006

Supplemental Instruction Leader - Supported an intensive summer program in which academically advanced incoming students completed the full calculus-based introductory physics sequence, PHYS 201–203, before beginning the regular academic year.

- Delivered focused problem-solving lectures between formal class meetings, working representative problems step by step and demonstrating efficient solution strategies.
- Helped students connect concepts across the compressed three-course sequence through diagrams, dimensional analysis, estimation, and physical interpretation of calculated results.
- Planned and led collaborative review sessions for groups of up to 12 students, integrating physics content with transferable learning strategies, addressing misconceptions, and modeling successful student behaviors.

Property Operations and Manual Labor**Doodlebug Farms**

Sep. 2024 - Present

Family Farm Operator - Establishes and operates a pre-production orchard with 72 American plum and Chickasaw plum trees.

- Manages seasonal tree care, grounds maintenance, supplies and equipment, and physically demanding outdoor work across the developing orchard.
- Plans staged expansion to approximately 216 trees, three times the current planting, while preparing the orchard to transition from establishment into harvest and sales.

Independent Contractor, Cleveland, Ohio

Summer 2003

Residential Painter and Maintenance Contractor - Completed home-improvement and property-maintenance projects.

- Prepared interior surfaces, applied paint, protected surrounding finishes, and maintained orderly work areas through project completion.

Independent Contractor, Cleveland, Ohio

Summer 2003

Concrete Repair Laborer - Repaired architectural concrete flooring and residential driveways for home-improvement and maintenance projects.

- Prepared damaged areas, placed and finished repair material, and completed job-site cleanup to restore worn concrete surfaces.

Crown Metal Finishing

Summer 2002

Production Laborer - Supported an industrial chrome-plating operation through hands-on material handling and line preparation.

- Loaded metal components onto production racks for machine-applied chrome plating, arranging parts for consistent processing and efficient line flow.

- Sustained production pace during repetitive manual work and followed shop-floor handling and safety procedures around plating equipment.

Gravity Games

Summer 2002

Event Operations Crew - Supported venue sanitation and waste-handling operations during a large public sporting event.

- Collected and disposed of waste and recycling throughout the venue and removed litter from high-traffic areas during and after the event.

FedEx

Winter 2002

Package Handler and Loader - Performed time-sensitive loading and material-handling work in a high-volume logistics environment.

- Completed physically demanding assignments involving repetitive lifting, rapid material movement, and coordination with other handlers.

Sales and Customer Service**Target**

Winter 2010

Sales Associate - Processed customer purchases and cash and credit transactions in a high-volume retail environment.

- Assisted customers, stocked merchandise, supported inventory activities, and helped resolve transaction concerns.

FranklinCovey

Jun. 2005 - Apr. 2006

Productivity Consultant - Advised customers on planning products, workshops, and productivity needs using detailed product knowledge.

- Matched products and services to individual customer goals and supported cash and credit transactions.

Dillard's

Aug. 2004 - Jun. 2005

Sales Associate - Helped customers evaluate merchandise and complete purchases.

- Supported merchandising, stocking, opening and closing activities, and customer-service issue resolution.

DialAmerica Marketing

Winter 2002

Marketing Representative - Supported telephone-based customer outreach and sales operations.

- Conducted customer outreach and sales conversations by telephone.

Food Service and Hospitality**Lincoln Park Pub**

Aug. 2013 - Mar. 2014

Cook - Supported restaurant kitchen operations in a fast-paced service environment.

- Prepared food and coordinated kitchen work to support accurate, timely meal service.

Aladdin's Eatery

Summer 2002

Dishwasher - Supported dish-room sanitation and restaurant kitchen operations.

- Maintained clean dishes, utensils, equipment, and work areas throughout restaurant service.

Barnacle Bill's Lobster House

Summer 2001

Cook - Supported kitchen production in a fast-paced restaurant environment.

- Prepared food and helped maintain timely meal production during restaurant service.

Cravings Cafe

Winter 2001

Barista - Provided beverage preparation and counter service in a cafe environment.

- Prepared coffee and espresso beverages, served customers, replenished supplies, and maintained the service area.

Pier W

Summer 2000

Cook - Supported food preparation and restaurant kitchen operations.

- Assisted with food preparation, kitchen cleaning, and meal-service activities.

Take 5 Coffee Company

Winter 2000

Barista - Supported day-to-day coffee-shop and counter-service operations.

- Prepared coffee and espresso beverages, stocked supplies, and completed opening and closing activities.

Credentials and Continuing Education**Professional Certificates**

Fundamentals of Visualization with Tableau - University of California, Davis

Jun. 2023

Credential ID M3VVMS4BTXWU.

IBM Data Science Professional Certificate

2022

Certificate of Specialization, Data Science - Johns Hopkins University

2015

Additional Coursework

2012 - 2014

Windows Forms; quantitative finance; information risk; computer networks; computational data analysis; AI planning.

IBM Digital Credentials

IBM Digital Credentials and Coursework

Nov. 2017 - Feb. 2022

Completed 43 distinct IBM credentials spanning data science, machine learning, data engineering, cloud, product management, design thinking, security, blockchain, and quantum computing.

- **Data science and machine learning:** IBM AI Associate Data Scientist; Data Science Foundations, Levels 1–2; Data Science Methodologies; Python, R, Scala, TensorFlow, GPU-accelerated deep learning, data visualization, and machine-learning coursework.

- **Data engineering and big data:** Hadoop Foundations and Programming; Hadoop Data Access; Spark; Apache Kafka data pipelines; Big Data Foundations, Levels 1–2; Data Science Tools.
- **Cloud and security:** IBM Cloud Private Infrastructure and Architecture; IBM Cloud Private Foundation Technology; Cloud Core; Cloud Service Management and Operations; Think Like a Hacker; Watson and Cloud Foundations.
- **Product, Agile, and design:** Agile Explorer; AgileTA Explorer; Product Manager curriculum covering product definition, delivery, business management, and scaling; Enterprise Design Thinking Practitioner and Co-Creator.
- **Emerging technologies:** Blockchain Essentials and Blockchain Essentials V2; Quantum Conversations; Watson Visual Recognition; Cognitive Practitioner.

Salesforce and Tableau Continuing Education**Salesforce Trailhead Coursework Completed at Arkos Health**

2023 - 2025

Reached Trailblazer Mountaineer rank with 18,725 points and 40 badges.

- **Tableau and data literacy:** Tableau Desktop, Tableau Cloud, Tableau Prep, visualization, data structures and field types, filtering and sorting, aggregation and granularity, data connections, collaboration, and presentation.
- **Salesforce administration and CRM:** Salesforce CRM, user management and engagement, Lightning Experience, formulas and validations, data modeling and management, and reports and dashboards.
- **Analytics, AI, and security:** CRM Analytics, correlation and regression, data-analytics fundamentals, natural-language processing, generative and responsible AI, Agentforce, and data security.

Education**M.S. Physics, University of Cincinnati**

Dec. 2010

Astrophysics research focus. Completed the thesis *Massive Stellar Clusters in the Disk of the Milky Way Galaxy*, using near-infrared photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk star clusters.

B.F.A. Printmaking, Ohio University

Aug. 2004

Studio focus in intaglio, relief, and screenprint. Completed a senior thesis exhibition and faculty-endorsed Provost's Undergraduate Research Fund work adapting a low-toxic toner-transfer process to intaglio photo-etch printing.